Humoral Regulation of Blood Pressure

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Adapted from: Humoral Regulation of Blood Pressure

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Explain how blood pressure is regulated by humoral factors.
2. Describe the release, targets, and mechanisms of action of certain systemic agents that **lower blood pressure**: natriuretic peptides, nitric oxide, histamine, and bradykinin.
3. Describe the release, targets, and mechanisms of action of two agents that **increase blood pressure**: endothelin and vasopressin.

CASE CONNECTION

MC has started taking a pre-workout powdered supplement to enhance his weightlifting capabilities. The powder is a proprietary blend of caffeine, creatine, and nitric oxide boosters or precursor such as L-arginine. He visits you in your office to ask a question. “I want my workout to be as efficient as possible. My only concern is that about a half hour after I take the powder, my veins look dilated. After I lift weights, the blood vessels appear even bigger. Do you think it’s okay to keep using the powder?”

How will you explain the effects of nitric oxide to MC? Consider your answer as you read, and we’ll revisit MC at the end of the brick.

GO TO CONCLUSION ↓

What Is the Humoral Regulation of Blood Pressure (LO1)?

Regulation of blood pressure is critical to avoid the adverse effects of extremes of blood pressure, where low blood pressure results in low oxygen delivery to the organs (ischemia), and high blood pressure results in vascular damage, stroke, cardiac and renal disease).

Many different mechanisms and factors regulate blood pressure. In general, rapid changes are mediated by the autonomic nervous system (ANS), while slower, long-term regulation, or local control of blood flow is primarily influenced by **humoral factors** (factors in body fluids, typically blood). This brick focuses on circulating humoral (related to body fluids) factors that affect blood pressure. Some serve as normal control mechanisms; including the natriuretic peptides and vasopressin. Others are relevant in disease states, such as histamine, a cause of life-threatening anaphylactic shock. Other circulating hormones that increase blood pressure, such as adrenaline and the renin-angiotensin-aldosterone system (RAAS), are covered elsewhere.

Molecules That Lower Blood Pressure (LO2)?

Natriuretic Peptides– ANP and BNP

Natriuretic peptides are important normal regulators of blood pressure and blood volume. They achieve this by both urinary natriuresis (increasing urinary sodium excretion) and inducing arterial vasodilation, which lowers systemic vascular resistance (SVR). There are two main natriuretic peptides: atrial natriuretic peptide (ANP) and "brain" natriuretic peptide (BNP). Despite their names, don't be misled; the word 'brain' in brain natriuretic peptide is somewhat of a misnomer:

- **ANP** (atrial, or A-type natriuretic peptide) is named appropriately, because it is secreted from atrial cardiomyocytes when stretch is detected in the heart.
- **BNP** (B-type natriuretic peptide) is released primarily from ventricular cardiomyocytes in response to increased ventricular wall stretch. BNP was originally known as “brain natriuretic peptide.” The term persists because BNP was originally identified in porcine brain tissue, not because it is produced there in humans.

The urinary sodium loss (natriuresis) is the most important function of these peptides and occurs by several mechanisms:

- **Glomerular filtration rate (GFR):** An increase in GFR increases the filtration of sodium.
- **Transport inhibition:** The sodium transporters of the renal tubules are inhibited, resulting in a decrease in sodium absorption in those structures.
- **Counteracting the RAAS system:** A decrease in renin secretion leads to a decrease in the formation of angiotensin II and aldosterone (the renin–angiotensin–aldosterone system, or RAAS). Since aldosterone causes

sodium reabsorption, the natriuretic peptides are effective counteragents because they decrease sodium reabsorption in the renal tubules.

How would this increase in the urinary loss of sodium lead to blood pressure reduction? Simple: if there is more salt remaining in the urine, the osmotic pressure in the urinary filtrate increases, causing water to move from the cells into the filtrate. This movement leads to an increase in urine excretion (diuresis). This fluid loss decreases the blood volume, thus decreasing cardiac preload, cardiac output, and blood pressure. The mechanisms are summarized in Figure 1.

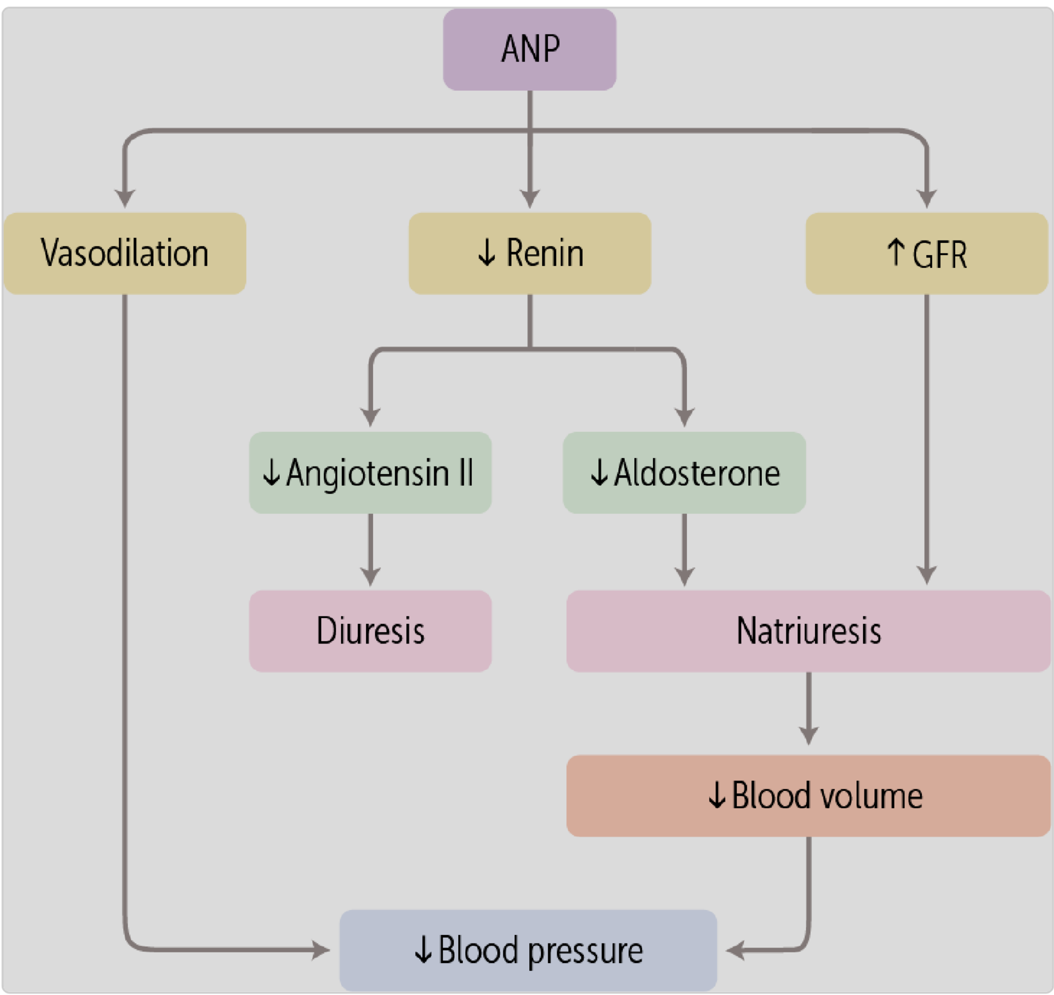

Figure 1 Effects of natriuretic peptides (ANP and BNP) on blood pressure and volume

They do this by increasing 1). the glomerular filtration rate (GFR), 2). by decreasing fluid reabsorption in the renal tubules, and 3). by decreasing renin secretion (leading to a reduction in aldosterone).

Nitric Oxide

Nitric oxide (NO) is a powerful vasodilating agent that's also important in everyday blood pressure regulation. It may be most familiar to you as the metabolic product of the drug nitroglycerin, used to dilate coronary arteries in patients with angina pectoris (ischemic heart disease). As an endogenous systemic vasodilator, it is released mainly from **vascular endothelial cells**, but another vasodilating isoform of NO is also made by neurons. NO circulates in the blood and binds to receptors on vascular smooth muscle cells to induce their relaxation, leading to vasodilation and BP reduction.

NO also acts locally, controlling regional blood flow in tissue beds. In its local role, it can dilate a vessel in response to shear forces on endothelial cells immediately adjacent to those cells that released it, dilating the vessel and directing more blood to the damaged region.

CLINICAL CORRELATION

Nitric oxide (NO) is approved as a drug for use in pediatric lung disease, acute respiratory distress syndrome (ARDS) and pulmonary hypertension. In these settings, it dilates the pulmonary vessels and increases blood flow to the damaged lungs.

Nitric oxide binds to vascular smooth muscle cells and induces them to relax, leading to vasodilation.

Histamine

Histamine is mostly a mediator of allergic reactions. However, it has many other functions, including increasing gastric acidity, alertness (antihistamines make you sleepy), pain and itchiness, and the permeability of vessels (e.g., nasal capillaries where antihistamines are used to treat your stuffy nose). Normally it is a mild vasodilator thought to work by dilating vascular beds to increase blood flow to sites exposed to allergens or inflammation. This allows other anti-inflammatory compounds to reach these exposed tissues. It can also be a powerful vasodilator in those disease states.

Histamine is well known for mediating allergic responses, inducing swelling, pain and itchiness. It also acts as a vasodilator by binding to receptors on vascular endothelium.

CLINICAL CORRELATION

Histamine in disease (anaphylaxis): In certain illnesses, dangerous levels of histamine can be released (e.g., an allergic response to peanuts). When this happens, the blood vessels dilate too much and the blood pressure drops too low, a condition called anaphylactic shock. This can cause hypoperfusion of organs.

Details of histamine biology are better left for discussions covering immunology. However, the main release of histamine is from **mast cells and basophils**, white blood cells that can release histamine granules. Then, histamine binds to its receptors located **on vascular endothelium**. This causes vasodilation and a reduction in blood pressure.

Mast cells and basophils are the main sources of histamine in peripheral tissues.

Bradykinin

Bradykinin is another vasodilating agent. It's primarily released during disease processes and promotes inflammation. Like histamine, bradykinin is mostly released by mast cells and basophils. Bradykinin synthesis is stimulated during inflammation, when specific proteases known as **kallikreins** are activated, allowing cleavage of bradykinin precursors (kininogens) into active bradykinin (Figure 2).

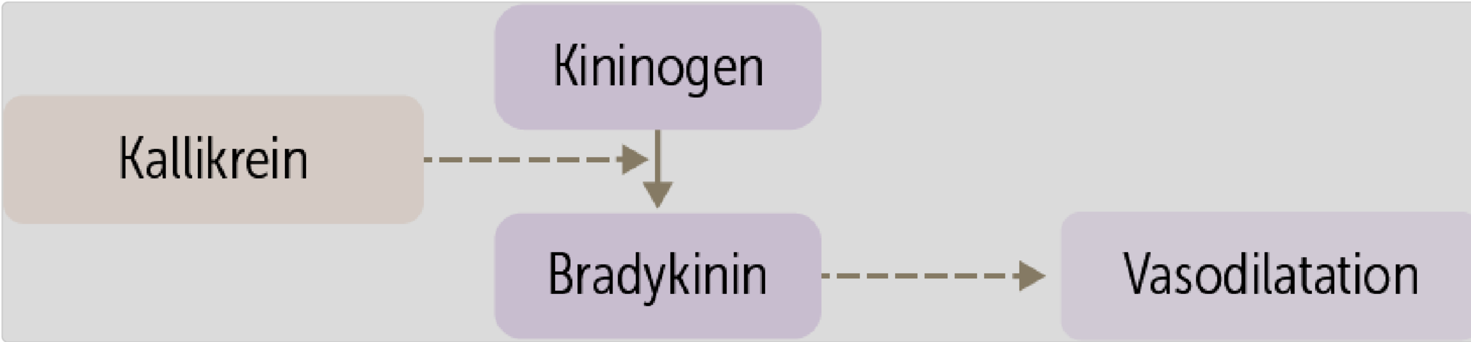

Figure 2 Bradykinin synthesis

There are two main types of bradykinin receptors. The bradykinin **B2 receptor is expressed widely in normal tissues**, including **vascular endothelial cells**. After bradykinin binds to cells, the cells secrete nitric oxide (NO), prostaglandins, and other molecules to cause vascular smooth muscle dilation. In contrast, the bradykinin B1 receptor is not normally expressed, only appearing in target cells after **trauma, burns, or inflammation**.

CLINICAL CORRELATION

Angioedema: An adverse clinical effect mediated by bradykinin is angioedema, or swelling of the lower skin layers. Angioedema is seen in some drug reactions, most famously with angiotensin-converting enzyme (ACE) inhibitors.

The bradykinin B2 receptor binds to endothelial cells, which then secrete nitric oxide, prostaglandins, and other molecules to cause the dilation of vascular smooth muscle.

Molecules That Increase Blood Pressure (LO3)

Unlike the molecules mentioned above, the following molecules stimulate the vascular smooth muscle cells to **constrict,** causing an increase in systemic vascular resistance and blood pressure. These vasoconstrictors are particularly useful in response to hypotension (such as after a severe hemorrhage), helping maintain blood pressure even when the blood volume drops.

Endothelin

The vasoconstrictor **endothelin,** as one might expect from its name, is synthesized and released from the vascular endothelium. Endothelin is released in response to shearing force from high blood flow, detected by the endothelial cells lining the blood vessels. Endothelin release is also stimulated by **angiotensin II and antidiuretic hormone (ADH).** Note that both of these substances lead to increases in blood pressure and/or blood volume, so they function synergistically with endothelin.

Endothelin binds to its receptors **(ET-A and ET-B),** inducing a **calcium-mediated** signal transduction pathway that allows **contraction** of smooth muscle fibers. This results in increased blood pressure.

CLINICAL CORRELATION

Pulmonary arterial hypertension: Endothelin is implicated in pulmonary arterial hypertension. In this disease, endothelin is abnormally elevated in the lung, leading to progressive vasoconstriction and pulmonary hypertension. This may lead to chronic shortness of breath due to impaired gas exchange. Endothelin antagonists and other vasodilating agents (eg, sildenafil) can be used to slow the progression of this disease.

Antidiuretic hormone (ADH)/Vasopressin

Antidiuretic hormone (ADH), or vasopressin, is best known for its effects in the kidney to reabsorb water. But it also causes systemic vasoconstriction. It activates arteriolar V_{1a} receptors on vascular smooth muscle cells and stimulates their contraction, constricting the arterioles and increasing blood pressure. This pairs nicely with its effect of expanding blood volume. (ADH stimulates renal reabsorption of water via the V_2 receptor on the principal cells of the collecting duct.) Both volume expansion and vasoconstriction will raise the blood pressure.

ADH is released by the posterior pituitary gland during states of high serum osmolality and low effective circulating volume (blood volume that effectively perfuses organs). Low effective circulating volume is seen in volume-depleted patients (i.e., dehydration) as well as patients with heart failure (low cardiac output) and liver or renal failure (high extracellular volume, but low blood volume because low serum albumin causes low serum oncotic pressure).

CLINICAL CORRELATION

Pharmacologic vasopressin-like agonists is used in intensive care units to increase the blood pressure in hypotensive patients as a complement to sympathetic vasoconstrictor agents like norepinephrine.

Vasopressin raises blood pressure by vasoconstriction and by stimulating renal water reabsorption to increase blood volume.

Summary of the Humoral Factors

For easy comparison, in Table 1, we've summarized the release, targets, and stimuli for each of the vasodilating and vasoconstricting molecules we've discussed.

Table 1 Comparison of humoral factors

Substance	Function	Released from	Target(s)	Upregulated by:
Histamine	Vasodilator Mediator of allergic response	Mast cells	Smooth muscle (H1 and H2 receptors)	Allergens

Substance	Function	Released from	Target(s)	Upregulated by:
Atrial natriuretic peptide (ANP)	Vasodilator Sodium excretion (urinary)	Atrial cardiomyocytes	Vasc. smooth muscle Kidneys	Atrial stretch
Brain natriuretic peptide (BNP)	Vasodilator Sodium excretion (urinary)	Ventricular cardiomyocytes	Vascular smooth muscle Kidneys	Ventricular stretch
Bradykinin	Vasodilator, inflammatory mediator	Cleaved from kininogen (from liver)	Vascular smooth muscle	Inflammation
Nitric oxide	Vasodilator	Vascular endothelium	Vascular smooth muscle	Inflammation, shearing force of blood flow
Endothelin	Vasoconstrictor	Vascular endothelium	Vascular smooth muscle	Angiotensin II, ADH, shearing force of blood flow
ADH/ Vasopressin	Vasoconstrictor Renal water reabsorption	Posterior pituitary	Vascular smooth muscle (V_{1a} receptors) Renal tubular cells (V2 receptors)	Low blood volume, high serum osmolality

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to MC, how do you explain the effects of nitric oxide to him?

You tell MC that the precursors in his powder that form nitric oxide are a powerful vasodilating agent. When the veins are enlarged to allow increased blood flow, more blood rushes toward the muscles. You inform him that whether these supplements really improve performance has not been proven. You also tell him that taking these supplements at low dosages should be okay if adverse effects do not occur, especially dizziness.

Summary

- Humoral (ie, related to body fluids) factors that circulate throughout the body are one of the many different mechanisms that regulate blood pressure.
- Natriuretic peptides such as ANP and BNP are vasodilators that promote the renal excretion of sodium, which leads to an increase in urine output and a reduction in blood volume.
- Nitric oxide is a vasodilator released by endothelial cells in response to local hemodynamic stress or inflammation; it relaxes vascular endothelial cells.
- Histamine is a potent vasodilator released from mast cells; it can be released during exposure to allergens and mediates anaphylactic shock.
- Bradykinin is a vasodilator made in the liver; it is released during inflammation and mediates the inflammatory response.
- Endothelin is a powerful vasoconstricting agent released from the vascular endothelium and is implicated in diseases such as pulmonary hypertension.
- ADH/ Vasopressin is a potent vasoconstrictor released from the posterior pituitary gland in states of low blood volume or high osmolality. It also stimulates renal water reabsorption.

Review Questions

1. A 28-year-old male presents to the emergency department in respiratory distress after experiencing multiple bee stings. He has a heart rate of 100 beats/min, blood pressure of 94/70 mm Hg, and respiratory rate of 24 breaths/min. Which of the following substances is most likely increased because of the bee stings?

A. Atrial natriuretic peptide

B. Bradykinin

C. Endothelin

D. Histamine

E. Vasopressin

Hide Explanation

The correct answer is histamine (D). The patient likely has an allergy to bee venom and is experiencing symptoms of anaphylactic shock (eg, hypotension, respiratory distress) due to histamine release from mast cells. Histamine causes vasodilation. As a result, this patient is at risk for tissue hypoxia and tissue hypoperfusion. Epinephrine increases cardiac output and vasoconstricts the surrounding vasculature; as a result, tissue perfusion increases, and the effects of the allergic reaction are countered. Incorrect answers: A). Atrial natriuretic peptide is a systemic vasodilator released when there is increased atrial stretch, as in heart failure, but not in allergic reactions. It also causes renal sodium excretion. B). Bradykinin is a vasodilator that causes angioedema in some drug reactions but is not as involved in allergic reactions. C). Endothelin is a vasoconstrictor, not a vasodilator, increased in pulmonary hypertension. E). Vasopressin is a vasoconstrictor, not a vasodilator. It also causes renal water reabsorption. It is not elevated in allergic reactions.

2. At your clinic, you see a patient with heart failure. Echocardiogram shows a dilated left atrium and a dilated left ventricle. The level of which of the following is most likely elevated in this patient because of the cardiac dilatation?

A. Bradykinin

B. Endothelin

C. Histamine

D. Natriuretic peptides

E. Nitric oxide

Hide Explanation

Correct answer D). The patient in this vignette has an enlarged heart, suggested by the dilation of the atrium and ventricle. When this occurs, the atrial and ventricular cardiomyocytes increase secretion of ANP and BNP, respectively. Neither atrial nor ventricular stretch directly increases in A). bradykinin, B). endothelin, C). histamine, or E). nitric oxide.

3. You are developing a pharmaceutical inhibitor of the endothelin receptor. Which of the following conditions might be a target for treatment by this drug?

A. Asthma

B. Hypotension

C. Angiodema (swelling under the skin)

D. Pulmonary arterial hypertension

Hide Explanation

Correct answer is E). Inhibiting endothelin receptors will inhibit vasoconstriction. Endothelin is involved in the pathogenesis of pulmonary arterial hypertension. As a result, inhibiting the action of endothelin would help treat this condition. There is no evidence that endothelin inhibitor can treat A). asthma, B). hypotension, C). angioedema, or D). hypotension.

4. A patient with low blood volume presents with increased vasoconstriction. Which hormone is most likely responsible for this vasoconstriction?

A. Bradykinin

B. Endothelin

C. Histamine

D. Nitric oxide

E. ADH (vasopressin)

Hide Explanation

Correct answer E). Antidiuretic hormone (ADH) or vasopressin causes water reabsorption in kidneys and vasoconstriction when blood volume is low. In correct answers: A). Bradykinin is a vasodilator, not a vasoconstrictor, and primarily functions during inflammation. B). While endothelin is a vasoconstrictor, it is not specifically released due to low blood volume like vasopressin. C). Histamine causes vasodilation and increased permeability, often related to allergic reactions. D). Nitric oxide is a vasodilator released by endothelial cells, not responsible for vasoconstriction.

How would you rate this Brick?

References

- Mast cell. *Wikipedia*. Accessed September 27, 2022. https://en.wikipedia.org/wiki/Mast_cell

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain how blood pressure is regulated by humoral factors.~~

02 ☑ ~~Describe the release, targets, and mechanisms of action of certain systemic agents that **lower blood pressure**: natriuretic peptides, nitric oxide, histamine, and bradykinin.~~

03 ☑ ~~Describe the release, targets, and mechanisms of action of two agents that **increase blood pressure**: endothelin and vasopressin.~~

Objective Confidence: 3/3